FACTORY REALLOCATION AND SHIPPING OPTIMIZATION

A Machine Learning-Based Scenario Analysis for Nassau Candy Distributor

Research Paper | Data Analytics & Machine Learning Project

Dataset size	10,194 order records
Models evaluated	Linear Regression, Random Forest, Gradient Boosting
Best model by R^2 and RMSE	Gradient Boosting Regressor
Scenario result	11 potential actual factory reallocation opportunities

Prepared from the supplied Nassau Candy Distributor dataset and completed notebook analysis. Predictions and scenario recommendations are exploratory and require validation before operational use.

Abstract

Shipping lead time is an important operational measure for distributors because delayed fulfilment can affect service quality, inventory planning, and operating efficiency. This project develops a machine-learning workflow to examine historical shipping patterns and compare alternative factory assignments for Nassau Candy Distributor. The dataset contains 10,194 records. Order and ship dates were parsed using the day-month-year format, and lead time was calculated as the difference between ship date and order date. Product, factory, region, shipping mode, order month, and order year were used as model inputs. Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor were evaluated using RMSE, MAE, and R^2. Gradient Boosting achieved the strongest R^2 (0.5334) and lowest RMSE (181.65 days) among the evaluated models, while Random Forest achieved the lowest MAE (172.72 days). A scenario simulation compared predictions across five factory options and generated 11 potential product reallocation opportunities. The mean model-predicted improvement across the recommendation table was 74.91 days (6.82%). These figures are model-based estimates, not observed savings. A key limitation is that Order Year dominates feature importance and alternative product-factory combinations are not represented historically. The results therefore support exploratory decision-making, but they do not establish causal factory effects or guarantee real-world lead-time or profit improvements.

Keywords: Shipping optimization, factory reallocation, lead-time prediction, Gradient Boosting, scenario analysis, data analytics.

1. Introduction

Distribution operations require coordination between products, production locations, destinations, and shipping methods. A fixed factory assignment may not always be the most suitable option for every product and destination profile. Data analytics can help an organization understand historical lead-time patterns and explore whether alternative assignments warrant further investigation. This project examines the potential use of supervised regression and scenario simulation to support factory-allocation decisions.

The analysis is framed as a decision-support prototype. The historical dataset records existing assignments and observed dates, but does not contain outcomes for the same product under every alternative factory. Consequently, the model can estimate outcomes for hypothetical configurations, but those estimates must not be interpreted as experimentally verified improvements.

2. Problem Statement

The project addresses the following question: can historical order information be used to predict shipping lead time and compare potential factory assignment scenarios in a structured way? The intended output is a ranked set of model-based recommendations that an operations team could review alongside capacity, distance, cost, service-level, and feasibility information.

3. Objectives

• Clean and validate date fields and calculate shipping lead time.

• Explore lead-time patterns across products, factories, regions, shipping modes, and order years.

• Train and compare three regression models using a consistent train-test split.

• Simulate each product against the available factory options and estimate predicted lead-time differences.

• Summarize recommendation coverage, a project-defined risk-prioritization proxy, and product gross-profit context.

• Document limitations so that model outputs are not confused with proven operational or financial outcomes.

4. Dataset and Preparation

The working dataset contains 10,194 records and includes order dates, ship dates, product names, shipping modes, destination regions, factory assignments, sales, cost, and gross profit. Product-to-factory mapping was applied using the project mapping rules. All 10,194 records were mapped to a factory, and the resulting modelling table had seven columns: Product Name, Factory, Region, Ship Mode, Order_Month, Order_Year, and Lead_Time_Days.

Date parsing required explicit handling because the source dates were in day-month-year format. The fields were parsed with `format="%d-%m-%Y"`. Lead time was calculated as `Ship Date - Order Date`, in days. After correction, there were no missing values in the order date, ship date, or calculated lead-time fields. The target variable had a mean of 1,320.84 days, a median of 1,274 days, and a range from 904 to 1,642 days.

The observed lead-time distribution forms several distinct bands rather than a smooth distribution. This pattern is important to interpretation and should be investigated with the dataset owner before operational decisions are made. The values were retained rather than arbitrarily altered.

Table 1. Dataset and preparation summary

Item	Result
Records	10,194
Factory mapping	10,194 mapped; 0 unmapped
Modelling inputs	Product, Factory, Region, Ship Mode, Order Month, Order Year
Target	Lead_Time_Days
Lead-time mean / median	1,320.84 / 1,274 days
Train / test split	8,155 / 2,039 records (80% / 20%)

5. Methodology

5.1 Exploratory Data Analysis

Descriptive statistics were calculated for lead time, factory, region, product, and order year. These summaries were used to understand record counts, central tendency, and variation. Factory and region averages were broadly similar, while order year showed a much stronger association with the lead-time bands in the available data. Products with very small record counts were treated cautiously because their summary statistics are less stable.

5.2 Feature Encoding and Model Training

The data was split into training and test sets using an 80/20 split with random_state=42. Categorical fields—Product Name, Factory, Region, and Ship Mode—were one-hot encoded with unknown categories ignored. Order Month and Order Year were retained as numeric inputs. The encoded feature matrix contained 30 columns. The target was Lead_Time_Days.

5.3 Regression Models and Evaluation

Three regression approaches were compared: Linear Regression as a baseline, Random Forest Regressor to model non-linear relationships, and Gradient Boosting Regressor to combine sequential weak learners. Model performance was assessed on the held-out test set using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). Lower RMSE and MAE indicate smaller prediction errors; a higher R^2 indicates that the model explains a larger share of target variation relative to a constant baseline.

5.4 Factory Scenario Simulation

For scenario analysis, each order was expanded across the five available factory options. The same encoded model features were used to generate a predicted lead time for each scenario. Predicted improvement was calculated by comparing each alternative scenario prediction with the current factory baseline. Recommendations where the selected scenario factory matched the current factory were excluded from the count of actual reallocation opportunities. Because alternative assignments are not observed in the historical records, the comparison is counterfactual.

6. Results

6.1 Model Performance

Model	RMSE (days)	MAE (days)	R^2
Linear Regression	182.45	181.43	0.5293
Random Forest Regressor	192.37	172.72	0.4767
Gradient Boosting Regressor	181.65	179.94	0.5334

Gradient Boosting performed best on two of the three reported metrics: it had the lowest RMSE (181.65 days) and highest R^2 (0.5334). Random Forest had the lowest MAE (172.72 days), so model selection depends on the metric prioritized. The overall explanatory power is moderate, and the error magnitudes remain substantial relative to many practical shipping windows.

6.2 Feature Importance and Interpretation

In the Gradient Boosting model, Order Year accounted for approximately 97.49% of reported feature importance, followed by Order Month at about 0.76%. The remaining features contributed much smaller shares. Exploratory analysis also showed that 2024 and 2025 records were distributed differently across the lead-time bands: 2024 records were in the short and medium groups, while 2025 records were in the medium and long groups. This strongly suggests that Order Year is acting as a proxy for temporal structure in the dataset. Feature importance is not proof of causality; the model therefore cannot be used to conclude that year itself or a particular factory causes a change in lead time.

6.3 Scenario Recommendations

The scenario simulation generated predictions for 10,194 records across five factory options (50,970 scenario rows). After excluding products whose best-scoring scenario remained their current factory, 11 potential actual reallocation opportunities remained. The project summary reports a mean predicted improvement of 74.91 days, equivalent to 6.82%, across its recommendation table. This average is a model output and should not be presented as an observed or guaranteed reduction.

Table 3. Illustrative top model-based reallocation scenarios

Product	Current factory	Scenario factory	Predicted improvement
Milk Chocolate	Wicked Choccy's	Secret Factory	130.11 days; 11.91%
Triple Dazzle Caramel	Wicked Choccy's	Secret Factory	130.11 days; 11.91%
Scrumdiddlyumptious	Lot's O' Nuts	Secret Factory	130.11 days; 11.79%
Nutty Crunch Surprise	Lot's O' Nuts	Secret Factory	117.81 days; 10.64%
Fizzy Lifting Drinks	Sugar Shack	Secret Factory	100.46 days; 9.42%

These options are candidates for further investigation rather than immediate transfer instructions. In particular, the model has not been trained on historical outcomes for alternative product-factory pairings, and the simulation does not account for factory capacity, transport distance, shipping cost, production constraints, or service-level commitments.

6.4 Risk Prioritization and Profitability Context

The project applied a rule-based risk-prioritization proxy based on predicted improvement percentage: six opportunities were labelled low risk, three medium risk, and two high risk. These labels are project-defined categories only; they are not validated operational risk estimates. Historical Gross Profit was summarized by product to add business context. For example, major Wonka Bar products had average gross profit values around 8–9 in the dataset, while some low-volume products showed much higher averages. Small sample sizes limit confidence in some product-level estimates. Since the dataset does not show profit changes caused by reassignment, direct incremental profit impact or profit stability cannot be quantified from the current analysis.

7. Key Performance Indicators

KPI	Project result	Interpretation
Selected model	Gradient Boosting	Best RMSE and R^2 among evaluated models
RMSE / MAE / R^2	181.65 / 179.94 / 0.5334	Moderate fit; errors remain substantial
Mean predicted improvement	74.91 days / 6.82%	Model estimate, not measured savings
Actual reassignment candidates	11	Scenarios with a different recommended factory
Risk proxy distribution	6 low / 3 medium / 2 high	Rule-based prioritization only
Profit impact	Not directly measurable	Gross Profit used for context, not causal impact

8. Limitations

• Historical data contains one existing factory assignment per product; alternative product-factory combinations lack historical observations.

• Order Year dominates model feature importance, and year aligns closely with the distinct lead-time bands. This may inflate apparent predictive usefulness for factory scenarios.

• The lead-time values are unusually large for ordinary shipping operations and cluster into narrow bands. Their business meaning should be verified with the data provider; this analysis retained the values as supplied after correct date parsing.

• The scenario model does not include factory capacity, geographic distance from destination, transportation cost, inventory, production throughput, or service-level constraints.

• The model uses a random train-test split rather than a time-based holdout. Further validation with temporal splits and leakage checks is needed.

• The reported 100% recommendation coverage means that the model generated positive predicted improvement for the products in the recommendation workflow; it does not mean 100% of recommendations will work in practice.

• Direct profit impact and a validated scenario-confidence score were not established by the available analysis.

9. Recommendations for Future Work

• Confirm the interpretation and units of Order Date, Ship Date, and Lead_Time_Days with the dataset owner.

• Validate the model using time-based cross-validation and compare against simple baselines.

• Collect observations for products produced or fulfilled by multiple factories, or conduct a controlled pilot before treating counterfactual predictions as evidence.

• Add destination distance, transport cost, factory capacity, production lead time, inventory, and service-level data.

• Build a decision dashboard that clearly separates observed historical metrics from model-generated scenario estimates.

• Measure actual outcomes after any pilot reallocation and compare realized lead time, cost, gross profit, and service levels with a baseline.

10. Conclusion

This project demonstrates a complete exploratory workflow for shipping lead-time analysis and factory scenario comparison: data preparation, exploratory analysis, regression modelling, model evaluation, scenario simulation, recommendation ranking, and profitability context. Gradient Boosting provided the strongest overall results by RMSE and R^2 among the tested models, and scenario analysis identified 11 potential factory reallocation candidates. However, the dominance of Order Year, the clustered lead-time values, and the absence of historical alternative-factory outcomes mean the current results should be treated as preliminary decision support. The next step is not automatic reassignment, but data validation, stronger model testing, and operational pilot evaluation.

In summary, the project provides a foundation for evidence-informed review of factory assignments, while making clear the distinction between historical observations, model predictions, and outcomes that would need to be validated in real operations.

References and Tools

• Project dataset and assignment brief: Nassau Candy Distributor dataset supplied for the project.

• Scikit-learn documentation — model selection, preprocessing, and regression metrics: https://scikit-learn.org/stable/

• Pandas documentation — data manipulation and date/time operations: https://pandas.pydata.org/docs/

• Matplotlib documentation — plotting and visualization: https://matplotlib.org/stable/

Note: This document reports the analysis and metrics available in the project notebook. It does not claim peer review, external validation, or measured business savings.